

Why Do Black Holes Exist?

From Einstein's Prediction to Penrose's Singularity Theorem

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Abstract—Black holes were long suspected to be mere artifacts of symmetry or idealization in Einstein's theory of gravitation. This paper traces the chain of reasoning that established their existence as a robust physical prediction. Starting from the Schwarzschild solution of 1916, we follow the development of stellar-collapse theory through the Chandrasekhar and Oppenheimer–Volkoff mass limits and the Oppenheimer–Snyder model of continued gravitational contraction. We then show how Penrose's 1965 singularity theorem removed the reliance on symmetry: once a trapped surface forms during the collapse of a sufficiently massive star, geodesic incompleteness follows from the causal structure of spacetime alone, regardless of the details of the collapse. The theoretical conclusion has since been confirmed by observations of Cygnus X-1, stellar orbits around Sagittarius A*, gravitational-wave detections of binary black-hole mergers, and horizon-scale imaging by the Event Horizon Telescope. The recognition of this work by the 2020 Nobel Prize in Physics is discussed.

Index Terms—black holes, general relativity, gravitational collapse, singularity theorems, Penrose theorem, event horizon, trapped surface

I. INTRODUCTION

Among the questions that general relativity forces upon physics, perhaps none is more striking than the one asked in the title of this paper: why do black holes exist? Einstein's field equations were published in November 1915 [1], and only weeks later Karl Schwarzschild wrote down the first exact solution describing the exterior field of a spherical mass [2]. That solution appeared to harbour pathological behaviour at a finite radius, and for nearly half a century most relativists regarded the associated “dark star” as a coordinate artifact rather than a realizable object. Einstein himself argued in 1939 that gravitational collapse to such a state could not occur in nature [3].

The resolution of this decades-long puzzle came from Roger Penrose, who proved in 1965 that the formation of singularities in gravitational collapse is not a consequence of idealized spherical symmetry but a *generic* and robust prediction of the theory [8]. For this discovery, described by the award citation as “the discovery that black hole formation is a robust prediction of the general theory of relativity,” Penrose received one half of the 2020 Nobel Prize in Physics, the other half

being awarded to Reinhard Genzel and Andrea Ghez for establishing that a supermassive compact object governs the orbits of stars at the centre of our Galaxy [20].

The purpose of this paper is to present, in a self-contained and compact form, the argument that answers the title question. Section II recalls the Schwarzschild solution and its physical interpretation. Section III reviews how the deaths of massive stars make gravitational collapse unavoidable. Section IV states and explains Penrose's singularity theorem. Section V summarizes the principal observational confirmations, and Section VI concludes with a discussion of open problems.

II. EINSTEIN'S PREDICTION: THE SCHWARZSCHILD SOLUTION

The dynamics of spacetime in general relativity are governed by Einstein's field equations,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (1)$$

which relate the Einstein tensor $G_{\mu\nu}$, built from the metric $g_{\mu\nu}$ and its first two derivatives, to the stress–energy tensor $T_{\mu\nu}$ of matter. For the vacuum exterior of a static, spherically symmetric body, with $\Lambda = 0$, Schwarzschild obtained the line element [2]

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (2)$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$ and

$$r_s = \frac{2GM}{c^2} \quad (3)$$

is the Schwarzschild radius. Heuristically, setting the Newtonian escape velocity equal to the speed of light reproduces exactly this radius, which already suggests that a sufficiently compact object would trap light itself.

Two features of (2) caused confusion. The metric component g_{rr} diverges at $r = r_s$, while curvature invariants remain perfectly regular there; by contrast, curvature scalars diverge at $r = 0$. We now understand $r = r_s$ as a coordinate singularity marking the event horizon, whereas $r = 0$ is a genuine curvature singularity. The decisive clarification came

from Finkelstein, who showed in 1958 that, in coordinates regular at the horizon, the surface $r = r_s$ acts as a one-way membrane: future-directed causal curves may cross it inward but never outward [7]. In modern language, the event horizon is the boundary of the causal past of future null infinity, and the Schwarzschild geometry describes a black hole.

It is worth emphasizing what the Schwarzschild solution by itself does *not* show. It is an eternal, exactly spherical vacuum solution, and nothing in it demonstrates that any physical process can produce such a geometry. This gap between exact mathematics and astrophysical reality is precisely what collapse theory and Penrose’s theorem were later to fill, against sustained skepticism [3].

III. GRAVITATIONAL COLLAPSE OF MASSIVE STARS

A main-sequence star maintains hydrostatic equilibrium because the pressure gradient of its hot plasma balances self-gravity. The pressure, in turn, is sustained by nuclear fusion. When nuclear fuel is exhausted, the core must contract, and the final state is determined by whether any known form of pressure can halt the contraction.

For cores below roughly $1.4 M_\odot$, the answer is yes: electron degeneracy pressure supports a white dwarf. This upper bound, the Chandrasekhar limit, was derived in 1931 [4] and follows from combining the Pauli exclusion principle with special relativity, which limits how much momentum a degenerate electron gas can supply.

For more massive remnants, electrons and protons combine into neutrons, and neutron degeneracy pressure can support a neutron star. The analogous mass limit for neutron cores was estimated by Oppenheimer and Volkoff in 1939 to be of order $0.7 M_\odot$ [5]; modern equations of state raise the value to approximately $2\text{--}3 M_\odot$, but it remains finite.

The decisive paper, however, was that of Oppenheimer and Snyder, also of 1939, “On Continued Gravitational Contraction” [6]. Modelling the collapsing core as a homogeneous sphere of pressureless dust, they matched an interior Friedmann–Lemaître geometry to the exterior Schwarzschild geometry and showed that, once contraction begins, nothing stops it. In the proper time of the infalling matter, the stellar surface crosses $r = r_s$ and reaches zero radius in finite time; to a distant observer, signals from the surface are progressively redshifted until the star appears to freeze and fade. Oppenheimer and Snyder concluded that the star “thus tends to close itself off from any communication with a distant observer; only its gravitational field persists.”

This is the astrophysical answer to why black holes exist: for a stellar remnant exceeding the maximum neutron-star mass, no equation of state permitted by known physics can supply sufficient pressure, and collapse proceeds without end. Table I summarizes the development.

Nevertheless, the Oppenheimer–Snyder calculation inherits an obvious weakness as evidence for real black holes: it assumes perfect spherical symmetry and exactly zero pressure. A critic could argue that angular momentum, pressure, inhomogeneity, or shock waves might cause a slightly perturbed

TABLE I
KEY MILESTONES IN THE THEORETICAL DEVELOPMENT OF THE
BLACK-HOLE CONCEPT.

Year	Milestone
1915	Einstein publishes the field equations of general relativity [1]
1916	Schwarzschild finds the first exact solution [2]
1931	Chandrasekhar derives the maximum mass of white dwarfs [4]
1939	Oppenheimer–Volkoff neutron-star limit [5]; Oppenheimer–Snyder collapse model [6]
1958	Finkelstein interprets r_s as a one-way horizon [7]
1965	Penrose proves the first general singularity theorem [8]
1968	Wheeler popularizes the term “black hole” [13]
2016	LIGO reports GW150914, a binary black-hole merger [18]
2019	Event Horizon Telescope images the shadow of M87* [19]
2020	Nobel Prize in Physics awarded to Penrose, Genzel, and Ghez [20]

collapse to rebound, or to fragment into pieces that escape. If so, the singular endpoint would be an artifact of idealization, much as many exact solutions are. Ruling out this objection required an entirely new kind of argument.

IV. PENROSE’S SINGULARITY THEOREM

A. New Tools: Causal Structure and Trapped Surfaces

Penrose’s breakthrough was methodological. Instead of solving the field equations explicitly, he applied global, topological methods to the causal structure of spacetime [8], [9]. Two concepts are central.

The first is the *convergence of light rays*. Consider a congruence of null geodesics with tangent field k^a and expansion θ , which measures the fractional rate of change of the congruence’s cross-sectional area. Along each geodesic, with affine parameter λ , the Raychaudhuri equation gives

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} - R_{ab}k^ak^b, \quad (4)$$

where σ_{ab} is the shear. Through the field equations (1), the last term is controlled by the local energy density measured by a light ray; under the null energy condition, $R_{ab}k^ak^b \geq 0$, so $d\theta/d\lambda \leq 0$. Gravity thus focuses light: once a congruence begins to converge, it converges ever faster, and the expansion diverges to $-\infty$ within a finite affine parameter.

The second concept is the *trapped surface*. Let S be a closed two-surface, such as a sphere surrounding the core of a collapsing star, and examine the two families of future-directed light rays orthogonal to S , one directed outward and one inward. In flat space the outward family expands. In sufficiently strong gravity, both families converge:

$$\theta_{\text{out}}|_S < 0 \quad \text{and} \quad \theta_{\text{in}}|_S < 0. \quad (5)$$

Condition (5) means that light itself is being dragged inward faster than it can move outward — the region is already inside a horizon in all but name. Trapped surfaces arise generically in the Oppenheimer–Snyder collapse once the stellar surface

passes inside $r = r_s$, and, crucially, the condition is open: small perturbations of the geometry preserve it. This is exactly the kind of statement that defeats the “small perturbations save the day” objection.

B. The Theorem

Penrose’s 1965 theorem can be stated as follows [8], [10].

Let a spacetime satisfy the null energy condition and contain a non-compact Cauchy surface (so that it is globally hyperbolic). If it contains a closed trapped surface, then it is null geodesically incomplete: at least one light ray terminates after a finite affine parameter.

The conclusion is the precise, definition-independent way of saying that the spacetime contains a singularity. The argument proceeds by contradiction. If all null geodesics were complete, the boundary of the future of the trapped surface, $\partial J^+(S)$, would be generated by null geodesics that never leave this boundary. By the focusing in (4), neighbouring generators would be forced to converge and enter the interior of $J^+(S)$, implying that $\partial J^+(S)$ is compact. But in a globally hyperbolic spacetime with a non-compact Cauchy surface, $\partial J^+(S)$ is necessarily non-compact. The contradiction establishes geodesic incompleteness.

The decisive point is that nowhere in the argument is any symmetry assumed. Spherical symmetry, homogeneity, and vanishing pressure play no role: the theorem applies to any spacetime, however irregular, that satisfies the energy condition and develops a trapped surface. That is the exact meaning of the Nobel citation — black hole formation is a *robust* prediction of general relativity [20].

C. Aftermath and Extensions

Penrose’s result triggered the systematic development of singularity theorems, culminating in the Hawking–Penrose theorems that replaced trapped surfaces with more general conditions and were applied to cosmology, showing that the Big Bang itself is a singular boundary of classical spacetime [11].

The theorem proves geodesic incompleteness rather than directly proving the existence of an event horizon. The link to black holes as observationally defined objects is supplied by the cosmic censorship conjecture, advanced by Penrose, according to which singularities formed from generic gravitational collapse are hidden behind horizons and cannot be seen from infinity [12]. Although still unproven in full generality, cosmic censorship is supported by extensive analytical and numerical evidence and is assumed in essentially all of black-hole astrophysics.

V. OBSERVATIONAL EVIDENCE

The theoretical edifice described above has been confirmed by four largely independent classes of observation, summarized in Table II.

TABLE II
SELECTED OBSERVATIONAL EVIDENCE FOR BLACK HOLES.

Object	Mass	Method	Key refs.
Cygnus X-1	$\sim 21 M_\odot$	X-ray binary dynamics	[14], [15]
Sgr A*	$4.1 \times 10^6 M_\odot$	Stellar orbits	[16], [17]
GW150914	$62 M_\odot$	Gravitational waves	[18]
	remnant		
M87*	$6.5 \times 10^9 M_\odot$	Shadow imaging	[19]

A. Stellar-Mass Black Holes in X-ray Binaries

Cygnus X-1, discovered as a bright X-ray source in 1964, was identified in 1972 as a binary system in which a visible supergiant star orbits an unseen companion [14]. The invisible object accretes matter from its companion, producing the X-rays, and dynamical mass measurements place it well above the maximum neutron-star mass. Modern analyses give a mass of roughly $21 M_\odot$ [15], leaving a black hole as the only viable interpretation.

B. The Supermassive Object at the Galactic Centre

Beginning in the 1990s, two groups tracked individual stars orbiting the radio source Sagittarius A* at the centre of the Milky Way. The star S2 completes an orbit of roughly 16 years, and Keplerian fits to the stellar orbits confine a mass of $(4.1 \pm 0.6) \times 10^6 M_\odot$ to a region smaller than the Solar System [16], [17]. No cluster of ordinary objects could survive at such density without collapsing, and the conclusion — a supermassive black hole — was recognized with the other half of the 2020 Nobel Prize [20].

C. Gravitational Waves from Black-Hole Mergers

On 14 September 2015, LIGO detected GW150914, the gravitational-wave signal of two black holes of about $36 M_\odot$ and $29 M_\odot$ inspiralling and merging into a final black hole of $62 M_\odot$, with roughly $3 M_\odot c^2$ radiated as gravitational waves [18]. The waveform agrees quantitatively with numerical solutions of Einstein’s equations for binary black-hole coalescence, including the final ringdown of the newly formed horizon. This is the most direct evidence that black holes not only exist but interact exactly as general relativity predicts.

D. Imaging the Shadow of a Black Hole

The Event Horizon Telescope, a global millimetre-wave interferometer, resolved horizon-scale emission around the supermassive object in M87 in 2019 [19]. The observed ring, of diameter consistent with the predicted shadow of a black hole of mass $6.5 \times 10^9 M_\odot$, images the photon capture region — the observational signature closest to the horizon itself. The same instrument imaged Sagittarius A* in 2022.

VI. DISCUSSION AND CONCLUSION

Why do black holes exist? The answer supplied by a century of work is that they exist because the known laws of physics contain no mechanism to prevent them. Once a stellar core exceeds the maximum mass supportable by degeneracy

pressure, contraction proceeds; once the collapsing matter is compressed within its own Schwarzschild radius, a trapped surface forms; and once a trapped surface forms, Penrose's theorem guarantees that spacetime cannot remain complete — a singularity is inevitable, and (as cosmic censorship asserts) it is clothed by an event horizon. The chain is unusually strong: each link is either a consequence of well-tested microscopic physics, a solution of the field equations, or a theorem requiring only generic energy and causality assumptions.

Two reservations should be noted. First, the singularity theorems demonstrate the breakdown of classical general relativity rather than the detailed structure of what replaces it; a quantum theory of gravity is expected to resolve the singularity itself, though not to remove the horizon for astrophysical black holes. Second, the information paradox raised by Hawking radiation indicates that black holes remain the sharpest probe we have of the tension between quantum mechanics and gravitation.

Far from undermining their existence, these open questions make black holes more valuable: they are simultaneously confirmed predictions of a 1915 theory and the laboratories in which that theory's successors will be tested.

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